



# Mark Scheme (Results)

Summer 2022

Pearson Edexcel International Advanced  
Subsidiary Level  
In Chemistry (WCH11)  
Paper 01: Structure, Bonding and Introduction  
to Organic Chemistry

## Section A (Multiple Choice)

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                       | Mark |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 1               | <p><b>The only correct answer is D</b> (isolated atoms, atoms in molecules and atoms in giant structures)</p> <p><i>A is not correct because elements also exist as isolated atoms and as molecules</i><br/> <i>B is not correct because elements also exist as isolated atoms</i><br/> <i>C is not correct because elements also exist as molecules</i></p> | 1    |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                | Mark |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 2               | <p><b>The only correct answer is C</b> (0.56 g)</p> <p><i>A is not correct because the <math>A_r</math> has been halved instead of doubled to give the <math>M_r</math></i><br/> <i>B is not correct because the <math>A_r</math> has been used instead of the <math>M_r</math></i><br/> <i>D is not correct because the <math>A_r</math> has been doubled twice to give the <math>M_r</math></i></p> | 1    |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Mark |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 3               | <p><b>The only correct answer is B</b> (<math>\text{MgO(s)} + 2\text{H}^+(\text{aq}) \rightarrow \text{Mg}^{2+}(\text{aq}) + \text{H}_2\text{O(l)}</math>)</p> <p><i>A is not correct because the sulfate spectator ions have not been eliminated</i><br/> <i>C is not correct because the magnesium oxide is involved in the change of state and the sulfate ion is not</i><br/> <i>D is not correct because the magnesium oxide is involved in the change of state</i></p> | 1    |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                               | Mark |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 4               | <p><b>The only correct answer is B (0.684)</b></p> <p><i>A is not correct because this is the number of moles of sodium chloride in 250 cm<sup>3</sup></i><br/> <i>C is not correct because this is the mass of sodium chloride in 250 cm<sup>3</sup></i><br/> <i>D is not correct because this is the concentration of sodium chloride in g dm<sup>-3</sup></i></p> | 1    |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                             | Mark |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 5               | <p><b>The only correct answer is D (CrCl<sub>3</sub>)</b></p> <p><i>A is not correct because this is the same as the ratio of silver ions to chloride ions</i><br/> <i>B is not correct because this is based on the ratio of volume of silver nitrate added to precipitate height</i><br/> <i>C is not correct because this is based on the ratio of precipitate height to volume of silver nitrate added</i></p> | 1    |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                        | Mark |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 6(a)            | <p><b>The only correct answer is A (20)</b></p> <p><i>B is not correct because this is the number of electrons in a Sc atom</i><br/> <i>C is not correct because this is the number of electrons in a Sc<sup>-</sup> ion</i><br/> <i>D is not correct because this is calculated by taking Z = 45 – 21 and then removing an electron.</i></p> | 1    |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                          | Mark |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 6(b)            | <p><b>The only correct answer is A (22.5)</b></p> <p><i>B is not correct because this is <math>0.5 \times (\text{mass number} + \text{atomic number})</math></i><br/> <i>C is not correct because this is the <math>m/z</math> value for an <math>\text{Sc}^+</math> ion</i><br/> <i>D is not correct because this is mass number <math>\times 2</math></i></p> | 1    |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                            | Mark |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 7               | <p><b>The only correct answer is D (<math>\text{I}(\text{g}) \rightarrow \text{I}^+(\text{g}) + \text{e}^-</math>)</b></p> <p><i>A is not correct because the iodine is molecular and in the solid state and forms 2 mol of ions</i><br/> <i>B is not correct because the iodine is molecular and forms 2 mol of ions</i><br/> <i>C is not correct because the iodine is molecular and in the solid state</i></p> | 1    |

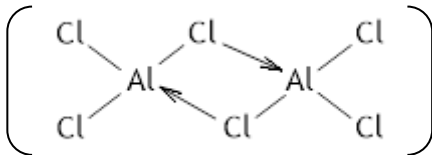
| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Mark |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 8               | <p><b>The only correct answer is C</b> <math>\left( \begin{array}{ccc} \boxed{\uparrow\downarrow} &amp; \boxed{\uparrow\downarrow} &amp; \boxed{\uparrow} \boxed{\uparrow} \boxed{\uparrow} \\ 1s &amp; 2s &amp; 2p \end{array} \right)</math></p> <p><i>A is not correct because the paired 1s and 2s electrons have parallel spin</i><br/> <i>B is not correct because the paired electrons have parallel spin and the 2p electrons do not have parallel spin</i><br/> <i>D is not correct because the unpaired 2p electrons do not have parallel spin</i></p> | 1    |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                       | Mark |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 9               | <p><b>The only correct answer is B</b> (8, 12, 5)</p> <p><i>A is not correct because the 4s electrons have been placed in the 3d subshell</i><br/> <i>C is not correct because the 4p subshell has been filled and 1 electron placed in 3d and 0 electron in 4s</i><br/> <i>D is not correct because the 4p subshell has been occupied before the 3d</i></p> | 1    |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Mark |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 10              | <p><b>The only correct answer is D</b> (attractive forces between oppositely charged ions, repulsive forces between like charged ions and some covalent bonding forces)</p> <p><i>A is not correct because the ions with the same charge will repel and there will be some covalency with <math>\text{Li}^+</math> and <math>\text{I}^-</math></i><br/> <i>B is not correct because there will be some covalency with <math>\text{Li}^+</math> and <math>\text{I}^-</math> ions</i><br/> <i>C is not correct because the ions with the same charge will repel</i></p> | 1    |

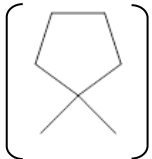
| Question number | Answer                                                                                                                                                                                                                                                                                                                                   | Mark |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 11(a)           | <p><b>The only correct answer is C</b> (M and N only)</p> <p><i>A is not correct because M is also a metal</i><br/> <i>B is not correct because L cannot be a metal because it is a poor conductor in the solid state</i><br/> <i>D is not correct because L cannot be a metal because it is a poor conductor in the solid state</i></p> | 1    |

| Question number | Answer                                                                                                                                                                                                                                                         | Mark |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 11(b)           | <p>The only correct answer is D (Q)</p> <p><i>A is not correct because L conducts in the liquid state</i><br/> <i>B is not correct because M conducts in the solid and liquid states</i><br/> <i>C is not correct because P does not dissolve in water</i></p> | 1    |

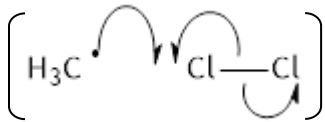
| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                      | Mark |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 12              | <p>The only correct answer is C</p>  <p><i>A is not correct because Al<sub>2</sub>Cl<sub>6</sub> is not ionic</i><br/> <i>B is not correct because the structure does not have a covalent bond between the aluminium atoms</i><br/> <i>D is not correct because the aluminium atoms have no lone pairs to donate in a dative bond</i></p> | 1    |

| Question number | Answer                                                                                                                                                                                                                                                                     | Mark |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 13              | <p>The only correct answer is A (the hazard is fixed but the risk varies)</p> <p><i>B is not correct because the hazard is fixed and the risk varies</i><br/> <i>C is not correct because the risk varies</i><br/> <i>D is not correct because the hazard is fixed</i></p> | 1    |

| Question number | Answer                                                                                                                                                                                                                                                                                       | Mark |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 14              | <p>The only correct answer is <b>B</b> (ions only)</p> <p><i>A is not correct because heterolytic fission only produces ions</i><br/> <i>C is not correct because heterolytic fission only produces ions</i><br/> <i>D is not correct because heterolytic fission only produces ions</i></p> | 1    |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                        | Mark |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 15              | <p>The only correct answer is <b>D</b> </p> <p><i>A is not correct because this compound has a molar mass of 100 g mol<sup>-1</sup></i><br/> <i>B is not correct because this compound has a molar mass of 114 g mol<sup>-1</sup></i><br/> <i>C is not correct because this compound would decolourise bromine water</i></p> | 1    |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                   | Mark |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 16              | <p>The only correct answer is <b>A</b> (ammonia)</p> <p><i>B is not correct because oxides of nitrogen are emitted in the combustion of alkane fuels</i><br/> <i>C is not correct because oxides of sulfur are emitted in the combustion of alkane fuels</i><br/> <i>D is not correct because unburnt hydrocarbons are emitted in the combustion of alkane fuels</i></p> | 1    |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                         | Mark |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 17              | <p>The only correct answer is B</p>  <p><i>A is not correct because two radicals reacting occurs in termination</i><br/> <i>C is not correct because a methyl radical reacting with methane would re-form the reactants</i><br/> <i>D is not correct because two molecules reacting together directly does not occur</i></p> | 1    |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Mark |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 18              | <p>The only correct answer is C (<i>E</i>-2-chlorobut-2-ene)</p> <p><i>A is not correct because cis-trans is not the IUPAC systematic name but is correct non-IUPAC name</i><br/> <i>B is not correct because cis-trans is not the IUPAC systematic name and is an incorrect non-IUPAC name</i><br/> <i>D is not correct because Cl is the priority group on the right-hand carbon and is on the opposite side of the double bond to the CH<sub>3</sub> on the left-hand carbon</i></p> | 1    |

**Total for Section A = 20 marks**



## Section B

| Question number | Answer                                                                                                                                                            | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                                                                          | Mark |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 19(a)(i)        | <ul style="list-style-type: none"> <li>calculation of mass of iron and use of <math>A_r(\text{Fe})</math> (1)</li> <li>evaluation of moles of iron (1)</li> </ul> | <p>Example of calculation:</p> $\text{mol} = \frac{6.17 - 3.38}{55.8} = \frac{2.79}{55.8}$ <p>mol iron = <math>5 \times 10^{-2} / 0.05</math> (mol)</p> <p>Allow <math>A_r</math> of Fe = 56 when mol Fe = 0.0498<br/>Ignore SF</p> <p>Correct answer with some working scores 2</p> <p>Use of incorrect mass for 1 mark TE<br/>e.g. 6.17g gives 0.11(057)<br/>3.38g gives 0.06(0057)</p> <p>Dividing 2.79g by an incorrect <math>A_r</math> gets 1 mark</p> | 2    |

| Question number | Answer                                                                                                                                                          | Additional guidance                                                                                                                                                                                                                                           | Mark |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 19(a)(ii)       | <ul style="list-style-type: none"> <li>expression for concentration and substitute values (1)</li> <li>evaluation of moles of iron(III) chloride (1)</li> </ul> | <p>Example of calculation:</p> $\text{concentration} = \frac{\text{mol}}{\text{vol in dm}^3}$ $0.500 = \frac{\text{mol}}{200/1000}$ $\text{mol} = 0.500 \times 200/1000$ $= 0.1 \text{ (mol)}$ <p>Correct answer with some working scores 2<br/>Ignore SF</p> | 2    |

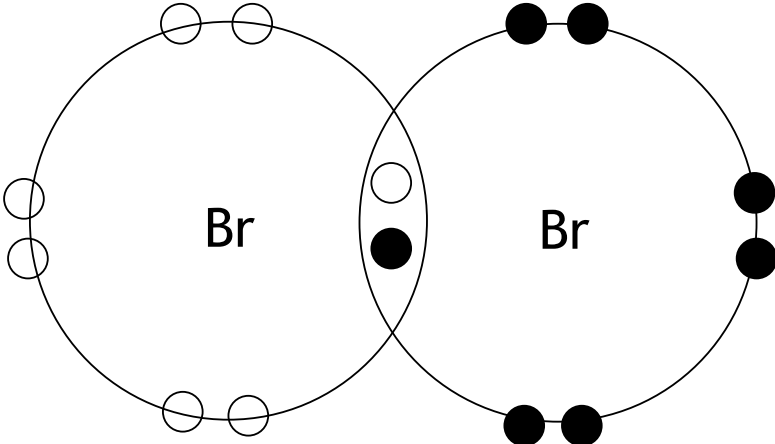
| Question number | Answer                                                                                                                                                    | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Mark |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 19(a)(iii)      | <ul style="list-style-type: none"> <li>calculation of whole number ratio (1)</li> <li>ionic equation (1)</li> <li>all three states correct (1)</li> </ul> | <p>Example of calculation:</p> <p>0.1 mol Fe<sup>3+</sup> (or FeCl<sub>3</sub>) reacts with 0.05 mol Fe<br/>so 2 mol Fe<sup>3+</sup> (or FeCl<sub>3</sub>) reacts with 1 mol Fe<br/>(0.1 ÷ 0.05 or 1 : 0.5 is enough for M1, not 2:1 alone)<br/>TE on incorrect i &amp; ii e.g. 1:1</p> <p>Fe(s) + 2Fe<sup>3+</sup>(aq) → 3Fe<sup>2+</sup>(aq)<br/>Allow multiples</p> <p>Allow correct states on species in the reaction<br/>Allow correct states on compounds, Cl<sup>-</sup> must be (aq)<br/>M2 and M3 standalone marks</p> <p>Comment:<br/>If no working shown max 2 marks</p> | 3    |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                  | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Mark |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 19(b)           | <ul style="list-style-type: none"> <li>• calculation of relative formula mass of <math>\text{FeCl}_2 \cdot x\text{H}_2\text{O}</math> (1)</li> <li>• calculation of relative formula mass of <math>\text{FeCl}_2</math> (1)</li> <li>• calculation of moles of <math>\text{H}_2\text{O}</math> in crystals (1)</li> <li>• calculation of moles of <math>\text{H}_2\text{O}</math> per mole of <math>\text{FeCl}_2</math> (1)</li> </ul> | <p>Example of calculation:</p> $\text{RFM}(1) = \frac{55.8}{28.1} \times 100 = 198.58$ $\text{RFM}(2) = 55.8 + 2 \times 35.5 = 126.8$ $\text{RFM}(1) - \text{RFM}(2) = 71.777$ $71.777 / 18 = 3.988 = 4$ <p>so <math>x = 4</math><br/> M4 is only awarded for final answer of 4</p> <p>Allow <math>A_r</math> of Fe = 56<br/> Correct answer with some appropriate working scores 4</p> <p>Ignore SF<br/> <u>Alternative:</u><br/> M1 mol Fe (in 100g hydrated salt) = <math>28.1/55.8 = 0.50358</math><br/> M2 mass of Cl (in 100g hydrated salt) = <math>0.50384 \times 2 \times 35.5 = 35.75(448)\text{g}</math><br/> M3 mass of <math>\text{H}_2\text{O}</math> (in 100g hydrated salt) = <math>100 - 28.1 - 35.75 = 36.15\text{g}</math><br/> moles <math>\text{H}_2\text{O} = 36.15/18 = 2.008</math><br/> M4 Ratio <math>\text{FeCl}_2 : \text{H}_2\text{O} = 0.5(0384):2(.008) = 1:4</math>, so <math>x = 4</math></p> <p><math>(55.8)/(55.8 + 71 + 18x) = 0.281</math> is another valid way of getting to <math>x = 4</math></p> | 4    |

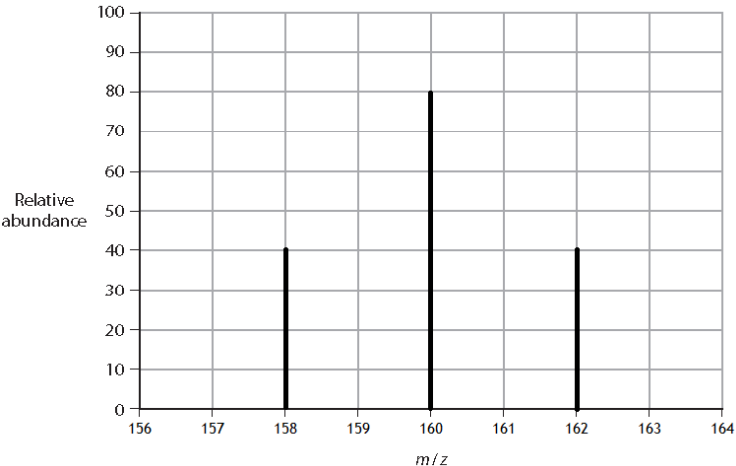
(Total for Question 19 = 11 marks)

| Question number | Answer                                                                                                                                                    | Additional guidance                                                                                                                                                                                                                                                                                                                              | Mark |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 20(a)           | <p>An answer that makes reference to</p> <ul style="list-style-type: none"> <li>(atoms with the) same atomic number and different mass numbers</li> </ul> | <p>Accept<br/>proton number for atomic number<br/>or<br/>same number of protons but different numbers of neutrons<br/>Allow<br/>bromine-79 has 35 protons &amp; 44 neutrons<br/><b>and</b><br/>bromine-81 has 35 protons &amp; 46 neutrons</p> <p>Allow "an atom" or "element"<br/>Do not award molecule<br/>Ignore same number of electrons</p> | 1    |

| Question number | Answer                                                                                                                                                                                                  | Additional guidance                                                                                                                                                                              | Mark    |          |           |    |    |    |    |    |    |   |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|----------|-----------|----|----|----|----|----|----|---|
| 20(b)           | <p>An answer that makes reference to:</p> <ul style="list-style-type: none"><li>all subatomic particles correct for bromine-79 (1)</li><li>all subatomic particles correct for bromine-81 (1)</li></ul> | <table><tr><th>Protons</th><th>Neutrons</th><th>Electrons</th></tr><tr><td>35</td><td>44</td><td>35</td></tr><tr><td>35</td><td>46</td><td>35</td></tr></table> <p>Any four correct scores 1</p> | Protons | Neutrons | Electrons | 35 | 44 | 35 | 35 | 46 | 35 | 2 |
| Protons         | Neutrons                                                                                                                                                                                                | Electrons                                                                                                                                                                                        |         |          |           |    |    |    |    |    |    |   |
| 35              | 44                                                                                                                                                                                                      | 35                                                                                                                                                                                               |         |          |           |    |    |    |    |    |    |   |
| 35              | 46                                                                                                                                                                                                      | 35                                                                                                                                                                                               |         |          |           |    |    |    |    |    |    |   |

| Question number | Answer                                                                                                                                                                                   | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                                                                  | Mark |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 20(c)(i)        | <p>A diagram showing:</p> <ul style="list-style-type: none"> <li>one shared pair of electrons (1)</li> <li>six non-bonding electrons on <b>both</b> atoms in the molecule (1)</li> </ul> | <p>Example of diagram:</p>  <p>Accept shared electrons on circles between the atoms<br/> Accept omission of circles or chemical symbols<br/> Allow any symbols for the electrons or elements (even if incorrect)<br/> Allow non-bonding electrons to be unpaired<br/> Allow horizontal sharing of bond pair<br/> Ignore horizontal line representing the bond</p> | 2    |

| Question number | Answer                                                                                                                                                                                                                                   | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Mark |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 20(c)(ii)       | <p>An answer that makes reference to:</p> <ul style="list-style-type: none"> <li>(a beam of) high energy electrons striking the (gaseous) bromine (molecule) (1)</li> <li>equation for the formation of the molecular ion (1)</li> </ul> | <p>Both marks may be scored with a correct equation and any indication the <b>bombarding</b> electrons are high energy</p> <p>Allow 'high speed electrons'<br/> Allow electron gun<br/> Allow "fast moving"<br/> Allow bromine atoms</p> <p><math>\text{Br}_2 + \text{e}^{(-)} \rightarrow \text{Br}_2^+ + 2\text{e}^{(-)}</math><br/> Allow<br/> <math>\text{Br}_2 \rightarrow \text{Br}_2^+ + \text{e}^{(-)}</math><br/> <math>\text{Br}_2 - \text{e}^{(-)} \rightarrow \text{Br}_2^+</math></p> <p>Allow <math>^{81}\text{Br}^{81}\text{Br}</math> (etc) instead of <math>\text{Br}_2</math> on either side of the equation</p> <p>Ignore state symbols even if incorrect</p> <p>Do not award <math>\text{Br}_2 \rightarrow 2\text{Br}^+ + 2\text{e}^{(-)}</math> or <math>\frac{1}{2}\text{Br}_2 \rightarrow \text{Br}^+ + \text{e}^{(-)}</math></p> | 2    |

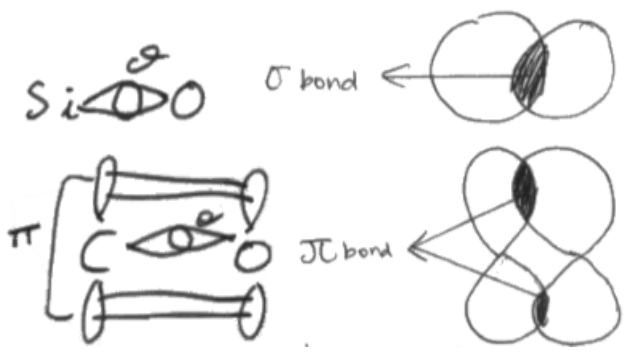
| Question number | Answer                                                                                                                                                                                                                                                 | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                           | Mark |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 20(c)(iii)      | <p>An answer on the grid showing:</p> <ul style="list-style-type: none"> <li>three peaks: at <math>m/z = 158</math>, <math>m/z = 160</math> and <math>m/z = 162</math> (1)</li> <li>peak heights in the ratio 1:2:1 (for 158: 160: 162) (1)</li> </ul> | <p>Example of grid:</p>  <p>Allow bars or "X" at the top of any lines<br/> Allow bars shown around the <math>m/z</math> values<br/> Do not award if a line is drawn from peak to peak</p> <p>Do not award if more than three peaks</p> <p>Allow any abundance values</p> <p>No TE on incorrect <math>m/z</math> values</p> | 2    |

| Question number | Answer                                                                                                                                                                                                                                | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                                             | Mark |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 20(d)           | <ul style="list-style-type: none"> <li>weighted mean mass expression (1)</li> <li>evaluation of relative atomic mass for Br (1)</li> <li>calculation of relative molecular mass for Br<sub>2</sub> corrected to 2 d.p. (1)</li> </ul> | <p>Example of calculation:</p> $A_r = \frac{79 \times 56.38 + 81 \times 43.62}{100}$ <p>= 79.87</p> <p>2 x 79.87 = 159.74 = 159.74</p> <p>Allow TE to M2 for values between 79 and 81<br/> Allow TE for M3 of double M2 value<br/> Ignore units even if incorrect</p> <p>Penalise rounding errors once only</p> <p>Correct answer to 2 d.p. scores (3)<br/> A <b>final</b> answer of 79.9 or 79.872 only<br/> scores 1 mark</p> | 3    |

(Total for Question 20 = 12 marks)



| Question number | Answer                                                                                                                                                                                                                                                                                                          | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                                                            | Mark |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 21(a)(i)        | <p>An answer that makes reference to the following points:</p> <ul style="list-style-type: none"> <li>the (strong electrostatic) <b>attraction</b> between the <b>shared</b> pair of <b>electrons</b> of the covalent bond (1)</li> <li>and the nuclei (of the silicon atom and the oxygen atom) (1)</li> </ul> | <p>Allow attraction between 2 shared electrons<br/>Ignore plurals<br/>Ignore bonding pair of electrons</p> <p>Allow "silicone" in place of silicon<br/>Allow references to protons instead of nuclei<br/>Do not award M2 for carbon atoms</p> <p>Ignore numbers of bonds</p> <p>Ignore references to giant/simple, double bonds, polar bonds, sigma bonds or orbital overlap</p> <p>Both marks may be scored by a clearly labelled diagram</p> | 2    |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                     | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                                    | Mark |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 21(a)(ii)       | <p>An answer that makes reference to the following points:</p> <p>Similarities:</p> <ul style="list-style-type: none"> <li>both molecules contain a <math>\sigma</math>-bond (1)</li> <li>description of end-on overlap (1)</li> </ul> <p>Difference:</p> <ul style="list-style-type: none"> <li>carbon dioxide (also) contains sideways overlap of orbitals / <math>\pi</math>-bond (with the oxygen atom) (1)</li> </ul> | <p>All marks may be scored by clearly labelled diagrams e.g.</p>  <p>Allow any indication that the <math>\sigma</math> overlap is between the atoms and the <math>\pi</math> overlap is above and below the plane of the atoms</p> <p>e.g. "end-on", "horizontal", "head on" for axial<br/>"sideways" or "parallel" for lateral</p> | 3    |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                       | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Mark |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 21(b)(i)        | <p>An explanation that makes reference to the following points:</p> <ul style="list-style-type: none"> <li>there are two sets of bonding electrons (and no lone pairs) about the carbon atom (1)</li> <li>which arrange to minimise repulsion (1)</li> <li>resulting in a linear shape / bond angle of <math>180^\circ</math> (1)</li> </ul> | <p>Any indication of two regions of electrons (this includes a correct diagram) e.g. <math>O=C=O</math><br/> Allow two double bonds<br/> Do not award MP1 for just "two bonding pairs" or just "4 bonding pairs"</p> <p>Allow maximum separation / to be as far apart as possible<br/> Do not award repulsion between atoms</p> <p>Accept bond angle labelled on a diagram<br/> Do not award linear with any angle other than <math>180^\circ</math></p> <p>Ignore references to symmetry<br/> Ignore references to lone pairs on oxygen</p> <p>All marks are independent<br/> No TE for any mark</p> | 3    |

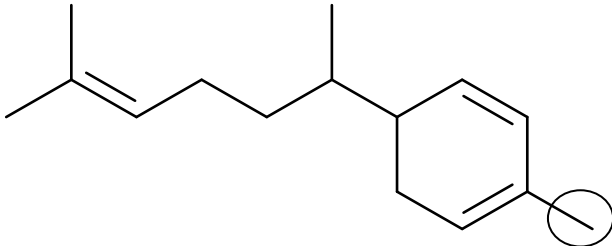
| Question number | Answer                                                                                                                                                                                                                                                                                                            | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                                          | Mark |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 21(b)(ii)       | <p>An explanation that makes reference to the following points:</p> <ul style="list-style-type: none"> <li>the carbon atom is slightly positive / <math>\delta+</math> <b>and</b> the oxygen atom slightly negative / <math>\delta-</math></li> <li>because oxygen is more electronegative than carbon</li> </ul> | <p>(1)</p> <p>Accept shown on a diagram e.g.</p> $\text{O}=\text{C}=\text{O}$ <p style="text-align: center;"><math>\delta+</math>   <math>\delta-</math></p> <p>Allow single C—O bond with dipole <math>\rightarrow</math></p> <p>Allow use of dipole symbol <math>\rightarrow</math></p> <p>Do not award full charges</p> <p>(1)</p> <p>Accept reverse argument</p> <p>Ignore “they have different electronegativities”</p> | 2    |

| Question number | Answer                                                                                                                                                                                                                                 | Additional guidance                                                                                                                                                  | Mark |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 21(b)(iii)      | <p>An answer that makes reference to the following point:</p> <ul style="list-style-type: none"> <li>the carbon dioxide molecule is not polar</li> </ul> <p><b>and</b></p> <p>because (it is a linear molecule) the dipoles cancel</p> | <p>(No TE on (b)(i))</p> <p>Allow the polar bonds cancel</p> <p>Allow dipoles balance</p> <p>Allow symmetrical molecule</p> <p>Do not award “the charges cancel”</p> | 1    |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                               | Additional guidance                                                                                                                                                                                                                                                    | Mark |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 21(c)(i)        | <p>An answer that makes reference to the following:</p> <ul style="list-style-type: none"> <li>each silicon atom has four silicon atoms in a <b>tetrahedral</b> arrangement as nearest silicon atom neighbours</li> </ul> <p>or</p> <p>each silicon is bonded to four oxygen atoms in a <b>tetrahedral</b> arrangement</p> <p>or</p> <p>each carbon is bonded to 4 carbon atoms in a <b>tetrahedral</b> arrangement (in diamond)</p> | <p>Allow "giant tetrahedral structure(s)"</p> <p>Do not award each silicon is tetrahedrally bonded to four other silicon atoms</p> <p>Ignore just "the silicon atoms are in a diamond structure"</p> <p>Do not award (simple) molecule(s) anywhere in the response</p> | 1    |

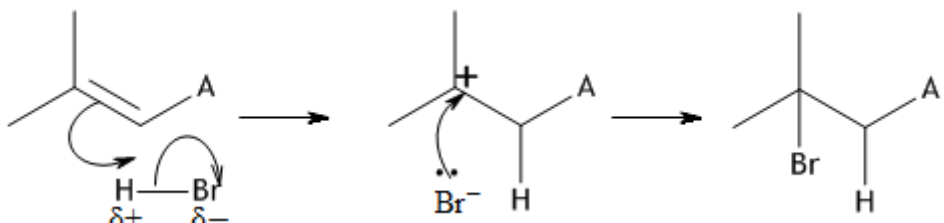
| Question number | Answer                                                                                                                                                                               | Additional guidance                                                                                                                                                                                                                                                                                          | Mark |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 21(c)(ii)       | <p>An answer that makes reference to the following point:</p> <ul style="list-style-type: none"> <li>There are fewer bonds (per atom) in the structure of silicon dioxide</li> </ul> | <p>Allow reverse argument</p> <p>Allow oxygen only forms two bonds</p> <p>Allow bond strength is an average</p> <p>Allow oxygen has lone pairs of electrons</p> <p>Ignore silicon-oxygen bond is polar</p> <p>Ignore references to sizes of atoms</p> <p>Do not award reference to intermolecular forces</p> | 1    |

(Total for Question 21 = 13 marks)

| Question number | Answer                                                                                                                                                                                 | Additional guidance                                                                                                                                                                                                                | Mark |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 22(a)           | <p>An answer that shows the following:</p> <ul style="list-style-type: none"> <li>any indication of the methyl group on the right-hand side of the structure being selected</li> </ul> | <p>Example of diagram:</p>  <p>Allow inclusion of the "2" carbon within the circle<br/>Do not award more than one group circled or selected</p> | 1    |

| Question number | Answer                                                                                                                                           | Additional guidance                                                                                                            | Mark |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|------|
| 22(b)           | <p>An answer that makes reference to the following:</p> <ul style="list-style-type: none"> <li>C<sub>15</sub></li> <li>H<sub>24</sub></li> </ul> | <p>(1) Allow H<sub>24</sub>C<sub>15</sub> / C<sub>15</sub>H<sub>24</sub><br/>(1) No TE on incorrect number of carbon atoms</p> | 2    |

| Question number | Answer                                                                                                                           | Additional guidance                                                                       | Mark |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|------|
| 22(c)(i)        | <p>An answer that makes reference to the following:</p> <ul style="list-style-type: none"> <li>electrophilic addition</li> </ul> | <p>Ignore 'heterolytic'<br/>Do not award 'free radical'<br/>Do not award substitution</p> | 1    |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Mark |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 22(c)(ii)       | <p>An answer that makes reference to the following:</p> <ul style="list-style-type: none"> <li>curly arrow from H—Br bond to Br atom or just beyond</li> </ul> <p><b>and</b></p> <p>dipole on H—Br (1)</p> <ul style="list-style-type: none"> <li>curly arrow from C=C to H or close by (1)</li> <li>structure of tertiary carbocation intermediate (1)</li> <li>curly arrow from <b>lone pair</b> on Br<sup>-</sup> to positively charged carbon atom (1)</li> </ul> | <p>Example of mechanism shown below</p>  <p>Addition of Br—Br loses M1 only</p> <p>Allow secondary carbocation (should be based on structure from paper)</p> <p>Do not award Br<sup>δ-</sup></p> <p>Do not penalise an incorrect product</p> <p>Allow structural formulae etc.</p> <p>Ignore omission of added hydrogen</p> <p>Ignore omission of A or substitution of A at any stage</p> <p>Ignore connectivity of CH<sub>3</sub> groups</p> <p>Ignore other lone pairs</p> | 4    |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                               | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                                   | Mark |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 22(c)(iii)      | <p>An explanation that makes reference to the following points:</p> <ul style="list-style-type: none"> <li>I is formed via a tertiary carbocation</li> </ul> <p><b>and</b></p> <p>II is formed via a secondary carbocation (1)</p><br><ul style="list-style-type: none"> <li>tertiary (carbocations) are more stable (than secondary) (1)</li> </ul> | <p>Assume "it" is I</p> <p>Must have carbocation at least once for M1<br/>"It is a tertiary carbocation" does not score M1</p><br><p>Allow secondary are more stable than primary for M2<br/>Allow tertiary is the most stable<br/>Allow reverse argument</p> <p>Marks are independent</p> <p>Ignore "tertiary cations have more alkyl groups"<br/>Allow "I is formed via a more stable intermediate" for M2 only</p> | 2    |

| Question number | Answer                                                                                                                | Additional guidance                          | Mark |
|-----------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------|------|
| 22(d)(i)        | <p>An answer that makes reference to the following:</p> <ul style="list-style-type: none"> <li>nickel / Ni</li> </ul> | <p>Accept platinum / Pt / palladium / Pd</p> | 1    |



| Question number | Answer                                                                                                                                                                                                                                                                                                                | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Mark |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 22(d)(ii)       | <ul style="list-style-type: none"> <li>rearrangement of ideal gas equation to make volume the subject (1)</li> <li>changing kPa to Pa <b>and</b> °C to K (1)</li> <li>substitution of values into IGE (including 2 x 3 mol of H<sub>2</sub>) (1)</li> <li>calculation of volume of hydrogen with units (1)</li> </ul> | <p>Example of calculation</p> $V = n \times R \times T \div p$ $p = 120 \times 1000 = 1.2 \times 10^5 \text{ (Pa)}$ <p><b>and</b></p> $T = 150 + 273 = 423 \text{ (K)}$ $V = 6 \times 8.31 \times 423 \div 1.2 \times 10^5 =$ $= 0.17576 \text{ m}^3$ <p>Accept 175.76 dm<sup>3</sup> / 175760 cm<sup>3</sup></p> <p>TE at each stage<br/>Ignore SF except 1 SF<br/>Correct answer <b>and</b> units with some working scores (4)</p> <p>Penalise incorrect rounding once only</p> | 4    |

(Total for Question 22 = 15 marks)

| Question number | Answer                                                                                                                                                                                                                    | Additional guidance                                                                                                                                | Mark |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 23(a)(i)        | <p>An answer that makes reference to one of the following:</p> <ul style="list-style-type: none"> <li>• biodegradation / putrefaction / decomposition</li> <li>• fermentation</li> <li>• anaerobic respiration</li> </ul> | <p>Ignore decay</p> <p>Allow any indication of a biological process e.g. 'bacterial action'</p> <p>Do not award "<b>thermal</b> decomposition"</p> | 1    |

| Question number | Answer                                                                                                                                                                                                                                        | Additional guidance                                                                                                                                                                                            | Mark |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 23(a)(ii)       | <p>An answer that makes reference to the following points:</p> <ul style="list-style-type: none"> <li>• climate change / global temperature (1)</li> <li>• methane / CH<sub>4</sub> <b>and</b> carbon dioxide / CO<sub>2</sub> (1)</li> </ul> | <p>Allow global warming</p> <p>Ignore greenhouse effect / increase in temperature</p> <p>Do not award ozone depletion / acid rain</p> <p>Do not award ammonia/NO<sub>x</sub>/SO<sub>x</sub>/H<sub>2</sub>S</p> | 2    |

| Question number | Answer                                                                                                                                                                                          | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Mark |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 23(a)(iii)      | <ul style="list-style-type: none"> <li>calculation of annual volume of carbon dioxide</li> <li>calculation of moles of carbon dioxide</li> <li>calculation of mass of carbon dioxide</li> </ul> | <p>(1) Example of calculation<br/> <math>= (45 \div 100) \times 365 \times 90000 \times 12.5 =</math><br/> <math>= 1.8478 \times 10^8 \text{ (dm}^3\text{)}</math></p> <p>(1) <math>= 1.8478 \times 10^8 \div 24(.0) = 7.6992 \times 10^6 \text{ (mol)}</math></p> <p>(1) <math>= 7.6992 \times 10^6 \times 44 = 3.3877 \times 10^8 \text{ (g)}</math><br/> <math>/ 3.3877 \times 10^2 \text{ tonnes} / 338.77 \text{ tonnes}</math></p> <p>Answers not in grams must have units for M3<br/> TE at each stage<br/> Allow 365.25 / 366 days<br/> Ignore SF except 1 SF<br/> Correct answer with some working scores (3)</p> | 3    |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                                           | Mark |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 23(b)           | <p>An answer that makes reference to any <b>two</b> of the following points:</p> <ul style="list-style-type: none"> <li>decreases quantity of waste / less space is needed (1)</li> <li>can be used (more easily) to generate electricity / produce heat (energy) (1)</li> <li>pollutants can be trapped more easily (1)</li> <li>transport costs lower (because sites can be sited near urban centres) (1)</li> <li>prevents release of <b>methane</b> into the atmosphere (1)</li> <li>high temperatures eliminate harmful bacteria / fungi (1)</li> <li>residue can be used in construction products (1)</li> <li>can deal with polymers/plastics/wastes that do not biodegrade (1)</li> <li>reduced risk of water / soil pollution (by leaching) (1)</li> </ul> | <p>Allow less land needed<br/>Allow no land needed <b>for waste</b><br/>Ignore no waste products</p> <p>Do not award generate energy<br/>Allow is a source of power</p> <p>Do not award incineration is less polluting / produces less CO<sub>2</sub> without referring to capture</p> <p>Allow residue can be used as fertiliser</p> <p>Ignore “reduces pollution”</p> <p>Ignore cost, appearance, time and contaminants</p> | 2    |

| Question number | Answer                                                                                                                                                                                                                | Additional guidance                                                                                                                                                           | Mark |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 23(c)           | <p>An answer that makes reference to <b>one</b> of the following points:</p> <ul style="list-style-type: none"> <li>resources conserved (by recycling) (1)</li> <li>less energy is used (in recycling) (1)</li> </ul> | <p>Allow reverse arguments</p> <p>Allow produces <b>less</b> toxic or greenhouse gases</p> <p>Do not award answers relating to cost</p> <p>Ignore renewable / sustainable</p> | 1    |

**(Total for Question 23 = 9 marks)**

**Total for Section B = 60 marks**

**Total for paper = 80 marks**

